Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2026

Course: B.Tech | *Time: 03 Hours*
Code: 155604 | *Subject: AI and Ethics* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
*(iv) Question No. **1** is compulsory.*

Q.1 Choose the correct option / answer the following (Any seven question only): **[2 x 7 = 14]**

(a) Which of the following best describes ethics in AI?
- (i) Rules for increasing computational speed
- (ii) Principles guiding responsible design and use of AI
- (iii) Methods for storing large datasets
- (iv) Techniques for reducing hardware cost

(b) Which is a possible psychological impact of AI on humans?
- (i) Increased dependence on automated systems
- (ii) Guaranteed improvement in memory
- (iii) Complete removal of workplace stress
- (iv) Elimination of human decision-making

(c) Which area is directly associated with the environmental impact of AI?
- (i) Energy consumption of computing systems
- (ii) Font selection in software
- (iii) Keyboard layout
- (iv) Screen resolution

(d) Which of the following is an ethical concern in autonomous vehicles?
- (i) Choice of paint colour
- (ii) Decision-making in safety-critical situations
- (iii) Size of the infotainment screen
- (iv) Brand of navigation display

(e) Transparency of autonomous systems primarily refers to
- (i) Keeping all AI decisions secret
- (ii) Making system behaviour and decisions understandable
- (iii) Removing all data from the system
- (iv) Allowing AI to operate without oversight

(f) Algorithmic bias may result in
- (i) Unfair or discriminatory outcomes
- (ii) Faster network connectivity
- (iii) Reduced data storage
- (iv) Improved battery capacity

(g) Data privacy processes are primarily concerned with
- (i) Protection and responsible handling of personal data
- (ii) Increasing processor speed
- (iii) Designing robot hardware
- (iv) Reducing software size

(h) Robotics taxonomy is used to
- (i) Classify different types and characteristics of robots
- (ii) Measure internet bandwidth
- (iii) Encrypt medical records
- (iv) Calculate AI training time

(i) Professional responsibility in robotics requires professionals to
- (i) Ignore societal consequences
- (ii) Consider ethical and social implications of their work
- (iii) Avoid all automation
- (iv) Use only open-source software

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(j) Which is an application area specifically identified as raising societal AI issues?
(i) Medicine (iii) Manual filing only
(ii) Handwriting with pen and paper (iv) Traditional painting

Q.2 (a) Explain the concepts of morality and ethics in the context of Artificial Intelligence. Discuss the relationship between AI and society. [7]

(b) Analyze the impact of AI on human psychology and trust. Illustrate your answer with suitable examples. [7]

Q.3 (a) Discuss the impact of Artificial Intelligence on the legal system and explain the major legal concerns arising from AI-based decision-making. [7]

(b) Examine the environmental impact of AI systems and discuss how responsible AI practices can reduce their impact on the planet. [7]

Q.4 (a) Discuss major international ethical initiatives in Artificial Intelligence. Explain the ethical harms and concerns that such initiatives seek to address. [7]

(b) Analyze the ethical issues involved in healthcare robots and autonomous vehicles, particularly with respect to safety, accountability, and human well-being. [7]

Q.5 (a) Discuss the ethical concerns associated with AI in warfare and weaponization. What responsibilities should be considered while deploying autonomous systems in such contexts? [7]

(b) Explain the model process for addressing ethical concerns during AI system design. Why should ethical considerations be incorporated throughout the design process? [7]

Q.6 (a) Explain the importance of transparency in autonomous systems. How can transparency contribute to accountability and trust in AI? [7]

(b) Discuss the data privacy process and algorithmic bias considerations that should be addressed while designing AI systems. [7]

Q.7 (a) Explain the need for an ontological standard for ethically driven robotics and automation systems. How can standards support responsible system design? [7]

(b) Discuss ethics and morality in robotics with reference to moral theories and ethics in science and technology. [7]

Q.8 (a) Discuss ethical issues in an ICT society and explain the importance of harmonization of principles, ethics, and professional responsibility in robotics. [7]

(b) Analyze societal issues concerning the application of Artificial Intelligence in medicine and its decision-making role in industries. [7]

Q.9 Write short note on ***any two*** of the following: [7 x 2 = 14]
(a) National and International Strategies on AI
(b) Robotics Taxonomy
(c) Ethical Harms and Concerns in AI
(d) AI, Society and Trust

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